

Coastal Wetland Restoration Notes

Page 1 - Site and objective

The fictional North Marsh teaching site covers 24 hectares of degraded coastal wetland.

The restoration objective is to increase native marsh vegetation and improve tidal water exchange while preserving a 3-hectare buffer for shorebird nesting.

The teaching plan treats vegetation recovery, tidal exchange, and bird disturbance as separate monitoring dimensions.

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Page 2 - Restoration actions

The first action is to remove two obsolete drainage barriers. The second is to regrade 6 hectares of compacted soil. The third is to plant three native species: smooth cordgrass, saltmeadow rush, and sea purslane.

Planting density is set at four plugs per square metre in the regraded zones.

No herbicide treatment is included in this fictional teaching plan.

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Page 3 - Monitoring design

Vegetation cover will be surveyed every two months at 18 fixed plots. Tidal water level will be recorded continuously by three loggers.

Shorebird observations will be conducted twice per month from designated observation points outside the nesting buffer.

The teaching team will compare Year 1 observations with a baseline survey completed before barrier removal.

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Page 4 - Interpretation limits

A single year of monitoring may show early ecological response but is not sufficient to establish long-term restoration success.

Storm events can temporarily alter water level and vegetation cover, so unusual events must be recorded alongside monitoring data.

The plan does not include fish-population sampling, water-chemistry analysis, or economic valuation.